Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Paper 5 Memory Rank Review Response

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Tue, Sep 22, 2026 at 12:12 AM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

Two things before the substance.

On the version. You are right, and the error is mine. I checked it against the file in my own examination set: the deposit I worked from is the
July-dated. None of Sara's checkable quotes appears in it.
The two reviews were therefore written against different texts, and my report raised as open several matters you had already repaired,
including both local errors and the terminological repair of the closure reading.
I should have pinned a revision number and a download date rather than a DOI, which on SSRN resolves to the current deposit;
my cover also carries a mistyped DOI.

On Remark 2.3. Your correction of my report is right, and I verified it rather than taking it on trust.
With the display I read, substituting Φ = Δ_V and using identity (5) gives δ_VΔ_V = 2(Δ_V(t,s,r) − Δ_V(t,s,q)), reproduced to $1.1 \cdot 10^{-16}$
on a random non-flat family over a five-point grid on three states, while δ_VΔ_V itself reaches 0.62.
My §3.1 asserted that display was correct and it was not. The corrected ordering satisfies δ_VΔ_V = 0 identically, your curvature formula (6)
checks to $5.6 \cdot 10^{-16}$ on the same family, and the analogy with d_A F_A = 0 and d_A² = F_A is exact;
answering my Question 16 in the negative is the right answer.

What the revision corrects. The distinction between exact realization and membership in the closed model set is now clear throughout:
h_d(V) = 0 certifies approximation by positive-block-mass lifts and not an exact lift, Remark 4.9 characterises the gap, and Example 4.10
witnesses it intrinsically.
The marginal-consistency obstruction is identified as one-sided, Definition 5.10 and Theorem 5.11 supply the stronger detector, and the text is
careful to distinguish their ordering as detectors from their strength as floors (a distinction I had not drawn).
The threshold rule of Remark 4.7 makes explicit what consistency of an integer-valued rank requires, which plug-in convergence of the profile
alone does not give. The treatment of the gauge class, the twisted coboundary, the projection geometry and "without a generator" is more precise;
general consequences of compactness and variational analysis are now labelled as such; and the quantum comparison is confined to a formal analogy.

Verification of the new material. None of it was before either report, so I re-derived it from the printed definitions.

- **Examples 8.2 and 8.3.** κ_c = 1 and κ_J = 1/4 in the first; κ_c = 0 and κ_J = 1/6 in the second, both by exact linear programming.
  In each case the optimal joint law has support exactly on the paths you print (the four of Example 8.2, and the six of Example 8.3 other than (1,2,1) and (2,1,2)).
- **Theorem 5.11(c).** No violation in 700 random families across n = 2, 3 and m = 2, 3, with both defects as exact linear programs.
  The ratio κ_c/(n²κ_J) reaches 1.0000, so n² is sharp and Example 8.2 attains it. A false start of my own, in case you recompute:
  κ_c by local minimisation rather than as the linear program of Definition 5.5 overestimates it and produced forty-two spurious violations before I did it properly.
- **Example 8.4.** min_M f(M) = 3.000000000 over the column-stochastic 2×2 matrices, attained only at M = J, so h_2^hom = √3 exactly;
  the family is flat, so h_2^inhom = 0; the Proposition 7.1 bound is √2 and is not attained.
- **Example 8.5.** The closed form holds to $4.9 \cdot 10^{-11}$ across four values of ε and every triple of a five-point grid.
- **Theorem 5.7(d) and Corollary 5.9.** The chain is sound: exact history realization gives P ⊆ κ(Θ⁺*{D_max}) ⊆ F*{D_max}, that set is closed,
  F_d ⊆ cl P for every d by part (b), and nesting closes it to equality above D_max, with D_max = 62 for n = 2, m

= 4.
The stabilization is a genuine strengthening of the repair I proposed, and the emptiness of the asymptotic liftability class is a better outcome
than the open question it replaces.

- Also checked: Example 4.10 analytically, including dist(V, κ(θ_ε)) = 2√3 ε on the three-point grid; Proposition 4.11, within its stated qualification
for homogeneous lifts with constant index gaps; and the repaired bound of Proposition 7.1, where the passage to a common Frobenius
lower bound is correct.
One caution: extracted from the PDF, the stacked fraction c/2 in that display reads as "2c" and briefly looked like a factor error;
it is not one in the typeset page.

On the withdrawal of Section 8. I agree with it. Upper bounds returned by optimization cannot establish positive distance to a lower-dimensional
model set or certify a minimal rank, and the archive and certificates existed to support a claim the tables could not carry.
Section 8 now gives exact witnesses and exact floors with hand-checkable arguments.
What the paper loses is worth recording somewhere: the class exchange rate, the most interesting observation in the old section,
survives only as Example 8.4 on a single family.

Limits to retain. Three boundaries belong in the examination record, as limits of the present results rather than unresolved errors in the revised proofs.

First, the withdrawal removed the uncertified rank claims but did not settle the underlying question.
The exact examples give upper bounds by witness for realizable families and positive floors for families the obstructions detect;
hey do not certify the minimal hidden dimension of an arbitrary non-flat process-induced family, where both obstructions vanish, and you say so plainly.
My Question 3 therefore stands as open rather than answered.

Second, when every optimizing joint law for κ_J has a zero one-time marginal, whether κ_J(V) = 0 implies membership in the closure of the
process-induced families is not established; Remark 5.12 states this, and it is the residual gap in Theorem 5.11(d).

Third, the compactified class that would convert all zero-distance statements into exact-realization statements is posed and not constructed,
so the closed-model reading is the operative one throughout.

Two requests.

1. **Attribution at the point of use.** A suggestion rather than a request, and perhaps just a difference in how the two of us read the revision: as it stands, the general acknowledgment is where the review contributions are recorded, and my own view is that one-line notes at Theorem 5.7(d), Definition 5.10 and Theorem 5.11, Example 4.10, Example 8.1 and Remark 3.6 would serve the paper better than a single thanks at the end. The development is yours in every case (the stabilization in particular goes well beyond the repair I proposed) so a footnote costs the paper nothing and makes the provenance legible to a later reader who comes to it without the correspondence. Entirely your call.
2. **Cross-reference with the coarse-graining paper.** Then new version of the manuscript cites the foundations, Koopman, convex-perturbation and selection-functional papers, but not *Memory Production under Coarse-Graining*, while that paper cites this one throughout as [21] and its Remark 2.7 states that its Proposition 2.6 *is* your Proposition 5.3 at d = n with exponent one and explicit constants. Three passages read differently once that is in view.
The paragraph after Proposition 5.3 says the exponent and C "are not constructive here", which is superseded at d = n.
Open Problem (ii) asks which contractions of Δ_V are comparable to h_n with exponent α = 1, which Proposition 2.6 answers for ‖Δ_V‖ itself.
And Open Problem (v) poses as open the decomposition of h_{n′}(ΛV) into a contractive image of the fine charge and "a production term with an identified zero set", which is that paper's balance law, with its Theorem 4.5 characterising the zero set and its Corollary 4.8 giving the charge-level bounds.
Both are September revisions, so chronology does not explain it. A problem posed as open in one paper of the portfolio and solved in another should be
recorded as solved in the first.

Once again, for the purposes of this paper's review, I regard the questions in my report as answered or correctly identified as open,
subject to the two requests above.

Best,
Ezequiel

[Quoted text hidden]